

Assignment 06

Question (1)

$$(1 - M_\infty^2) \phi_{xx} + \phi_{yy} = 0$$

Solution

Equation type $\rightarrow$ Linear second order PDE $\rightarrow a u_x^2 - 2b u_{xy} + c u_{yy} = f(x, y, u, u_x, u_y)$

by comparison $a = (1 - M_\infty^2)$ $-2b = 0$ $c = 1$ $f = 0$

$$b^2 - ac = 0 - (1 - M_\infty^2) \cdot 1 = M_\infty^2 - 1$$

If $M_\infty^2 > 1 \rightarrow$ hyperbolic

$M_\infty^2 = 1 \rightarrow$ parabolic

$M_\infty^2 < 1 \rightarrow$ elliptic

Question (2)

$$U_{xx} = X U_{yy}$$

Solution

(a) for Linear 2nd order PDE $\rightarrow a(x,y)U_{xx} - 2b(x,y)U_{xy} + c(x,y)U_{yy} = f(x,y,u_x,u_y,u)$

by Comparison $\rightarrow$ the equation is Linear

b)

$$a=1 \quad 2-2b=0 \quad c=-X \quad f=0$$

$$\therefore b^2 - ac = 0 - (1)*(-X) = X$$

If $X > 0 \rightarrow$ hyperbolic

$X = 0 \rightarrow$ parabolic

$X < 0 \rightarrow$ elliptic

(c)

Recall that,

$$\frac{dy}{dx} = \frac{b \pm \sqrt{b^2 - ac}}{a} = \frac{0 \pm \sqrt{0 - (1)*(-X)}}{1} = \pm \sqrt{X}$$

if $X > 0$

$$\therefore \frac{dy}{dx} = \sqrt{x} \rightarrow \int dy = \int \sqrt{x} dx \rightarrow y = \pm \frac{2}{3} x^{3/2} + C$$

if $X = 0$

$$\frac{dy}{dx} = 0 \rightarrow \int dy = \int 0 dx \rightarrow y = C$$

Question 13)

$$u_t + u u_x = 0$$

$$\text{Initial Condition} \rightarrow u(0, x) = \frac{1}{1+x^2}$$

Solution

(1), (2)

$$(1 \rightarrow u \rightarrow 0) \cdot (u_t \rightarrow u_x \rightarrow -1) = 0 \quad \& \quad du = u_t dt + u_x dx$$

$$(dt \rightarrow dx \rightarrow du) \cdot (u_t \rightarrow u_x \rightarrow -1) = 0$$

$$\therefore \frac{dt}{1} = \frac{dx}{u} = \frac{du}{0} = ds \rightarrow \frac{dt}{ds} = 1 \quad \& \quad \frac{dx}{ds} = u \quad \& \quad \frac{du}{ds} = 0$$

$$\therefore t = s + A \quad \& \quad x = us + B \quad \& \quad u = K$$

$$\text{Initially} \rightarrow s = 0 \quad \& \quad x = r \quad \& \quad t = 0 \quad \& \quad u(0, r) = \frac{1}{1+r^2}$$

$$\therefore A = 0 \quad \& \quad B = r \quad \& \quad K = u(0, x) = \frac{1}{1+r^2}$$

$$\therefore t = s \quad \& \quad x = \frac{1}{1+r^2} t + r \quad \& \quad u = \frac{1}{1+r^2}$$

$$(3) \quad \frac{\partial u}{\partial x} = \infty$$

$$\frac{\partial u}{\partial r} \frac{\partial r}{\partial x} = \infty$$

$$\therefore \frac{\frac{\partial u}{\partial r}}{\frac{\partial x}{\partial r}} = \infty \rightarrow \frac{\partial x}{\partial r} = 0$$

$$\frac{\partial x}{\partial r} = 1 - 2r + (1+r^2)^{-2} = 0$$

$$\therefore t_{sw} = \frac{(1+r^2)^2}{2r}$$

to get minimum time where SW happens

$$\frac{\partial t_{sw}}{\partial r} = 0$$

$$\therefore \frac{\partial t_{sw}}{\partial r} = \frac{1}{2} \left[r^{-1} * 2 * 2r(1+r^2)^2 + (1+r^2)^2 * -r^{-2} \right] = 0$$

$$\therefore 4 + 4r^2 - r^{-2} - r^2 = 3r^2 - r^{-2} + 2 = 0$$

$$\therefore 3r^4 + 2r^2 - 1 = 0$$

$$\therefore r^2 = \frac{-2 \pm \sqrt{4+12}}{6} = \frac{1}{\sqrt{3}} \rightarrow t_{sw \min} = \frac{8\sqrt{3}}{9} = 1.5396$$

Question (4)

(a) As A_x, A_y has the variables themselves so the system is quasi-linear

(b)

(i) $\lambda = u, u, \sqrt{\frac{u^2}{g}} + u, -\sqrt{\frac{u^2}{g}} + u \rightarrow \text{all real} \rightarrow \text{hyperbolic}$

(ii) $\lambda = v, v, \sqrt{\frac{v^2}{g}} + v, -\sqrt{\frac{v^2}{g}} + v \rightarrow \text{all real} \rightarrow \text{hyperbolic}$

(iii) $\lambda = \eta u + \epsilon v, \eta u + \epsilon v, \eta u + \epsilon v + \sqrt{(\eta^2 + \epsilon^2) \frac{u^2}{g}}, \eta u + \epsilon v - \sqrt{(\eta^2 + \epsilon^2) \frac{u^2}{g}} \rightarrow \text{all real} \rightarrow \text{hyperbolic}$

(c) $\lambda = v$